

03-3 | Lab 3: Introduction to Thematic Analysis

9/16/2026

Attempt 1



In Progress

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In this laboratory you will be performing a thematic analysis of your interview transcript and using it to identify needs and design goals. It is a long lab. You will have all of Friday, the weekend, and Monday to do it.

Part I: Open Coding (Recommendation: done by end of class Friday)

In the open coding step, you will go line by line and note things mentioned by your interviewee that you find significant in the context of your study. This step will allow you to go from text to concepts, providing an abstract, conceptual representation of your data. The concepts you identify are called “codes” and the process of marking your data for these codes is called “coding.” As you code your data, you will see that some of the concepts you identify reoccur and you start reusing the codes you created at the beginning of the coding process.

As a group, code each line in your Google Sheet with concepts in a separate column. Each line might need more than one code. In this case, use the second column of your spreadsheet for the primary code and add other codes in subsequent columns. Make sure that you choose codes that can be interpreted by others and make sense in the context of your data. Your codes should be simple, action-oriented, and avoid big-picture assumptions.

Part II: Thematic Clustering (Recommendation: done by end of class Friday)

Next, as a group work to cluster your codes into themes (distinct ideas that multiple codes fall under). It may be helpful to do this by writing codes on sticky notes and grouping them, or some other method. Once you've settled on a set of themes, add a new column to the right, and list the themes each line falls under, based on which codes that line is coded under. Finally, look at your themes and if necessary, cluster those themes into higher level subthemes in the exact same way. You can do this recursively as many times as necessary, but you should probably have a preference for a large number of shallow trees over a single very deep tree. Finally, draw this tree

in google slides or your favorite mind mapping software. As a reminder, make sure your category names are interpretable by others — avoid using abbreviations or shortcuts.

Part III: Writeup (Recommendation: done over the weekend)

In this phase, you will write up the results of your analysis. First, create a document (in overleaf).

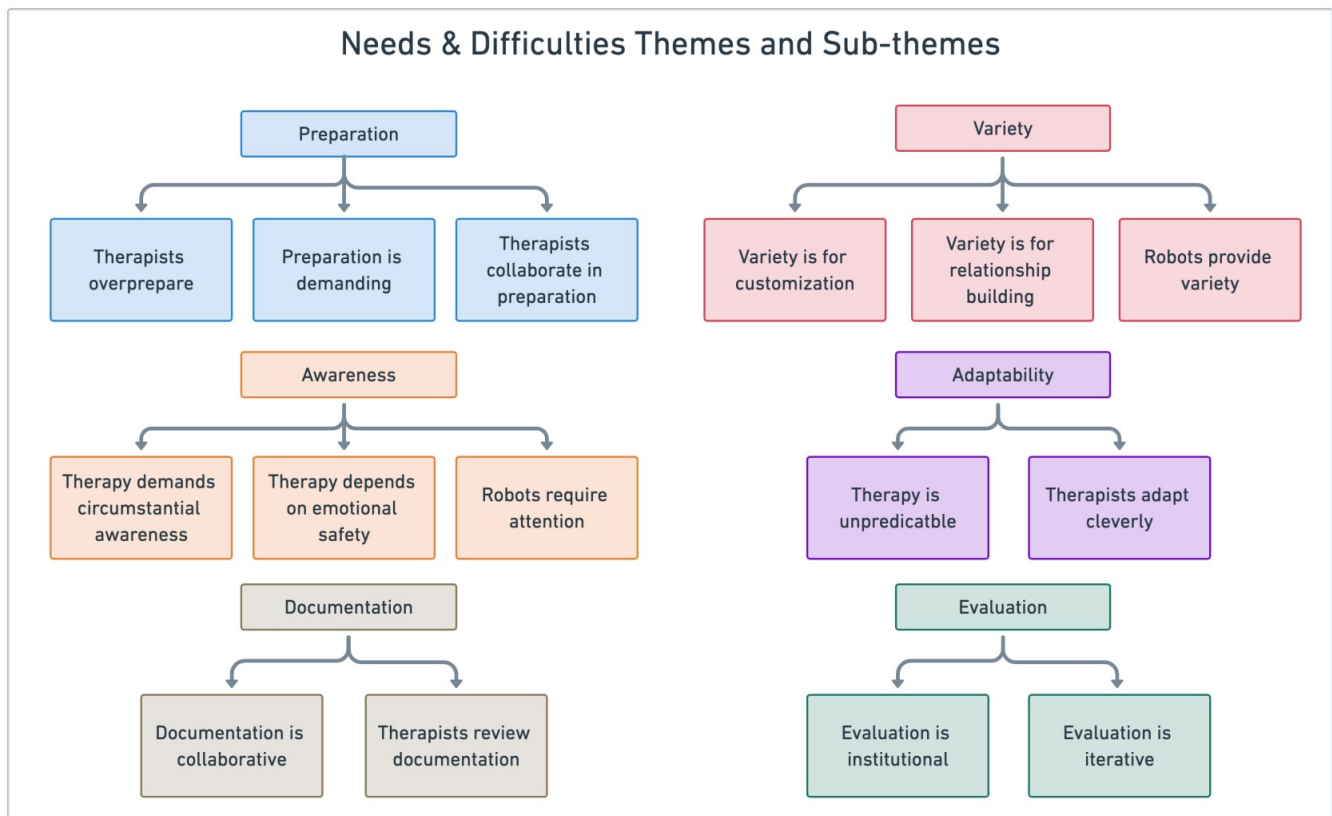
Methodology

Section 1 should be Methodology. Write a few paragraphs on who you interviewed (don't use their name) and why, your interview approach, and your analysis approach.

Results

Section 2 should be Results. This section will have a hierarchical structure. For each of your top-level themes, you will have a subsection. For each subtheme within that theme, you will have a subsubsection, or a paragraph (You probably shouldn't go deeper than three levels of nesting).

Consider for example this diagram of the themes from Saad Elbeleidy's paper "Centering Teleoperators in Robot-Child Interactions Reveals the Bigger Picture". Saad has six themes, each of which has three subthemes. So, Saad's results are divided into six subsections, each of which has three paragraphs within it.



Then, add your signposting. At the beginning of each section and subsection write a sentence that describes how the theme associated with that section is about to break down.

At the top of your results section this might look like a list of your high-level themes and their definitions:

We identified six themes that consistently emerged across our interviews (as visualized in Figure 4):

- (1) **Preparation:** therapists need to create or organize the resources they need for therapy.
- (2) **Variety:** therapists rely on a variety of content to deliver effective therapy.
- (3) **Awareness:** therapists maintain a high level of awareness in therapy to respond to their client and surroundings.
- (4) **Adaptability:** therapists adapt within each session to clients' needs in the moment.
- (5) **Documentation:** therapists rely on thorough documentation.
- (6) **Evaluation:** therapists continuously evaluate their clients to determine progress and upcoming interventions.

Below this list of themes, include a diagram like that shown above.

At the second level of your results, this will be more narrative. For example, if you have a theme "Children experience fear while at the doctor's office" with three subthemes (which are the names of the subsections within this section) "Children are afraid of needles", "Children are afraid of doctors", "Children are hesitant to share their fear", you might write "Our participants shared several ways that children's fear manifested at the doctor's office: (1) their fear of needles, (2) their fear of doctors, and (3) the ways they are hesitant to share their fear".

In Saad's paper, in the preparation subsection, it looks like this: "Effective preparation is necessary for effective therapy. Yet preparation is often omitted or disregarded in work focusing solely on the inner cycle of therapy sessions with children. In contrast, framing therapists as robot users naturally centers their preparation process. Our participants explained how demanding preparation can be, how the uncertainty of therapy leads them to over-prepare, and how introducing robots complicates their preparation process. In tension with these feelings, therapists also reported that they only feel prepared when they're over-prepared. When they use a robot, this need for over-preparation becomes even more salient, since preparation with a robot is more time consuming."

At the lowest level, where your themes are grounding down to your low-level codes, you will need to actually talk about your codes and provide quotes! Let's see how Saad handles this.

5.1.1 Preparation is demanding. Preparation is a fundamental part of therapy. To prepare, therapists read documentation about their clients' goals and previous sessions (see Section 5.5.1) to create or organize the appropriate interventions. When we discussed preparation with Holly (E1), they explained that *"you're preparing your lesson. ahead of time, based on what the goals would be for that session"*. Blanche (T2), a music and dance therapist, describing their preparation for sessions without a robot, says *"the first thing I do is set up my playlist"*³ Blanche (T2) says that when selecting music, *"the music really has to support the goals of the clients as well, so it is a big deal. I spend, a ridiculous amount of time nerding out about music"*. For Blanche (T2), it's clear that the preparation process is time-consuming and demanding. Blanche (T2) explains that *"For this week, maybe I know we want to work on self regulation ... so I'm going to, I know I'm planning to play a freeze dance game"*. Blanche (T2) shows us that preparation isn't just a one-time occurrence but instead changes every week or session. Holly (E1), a teacher with over 20 years of experience, mentioned that *"there's always preparation, no matter how experienced you are."* Preparing for therapy is clearly time-consuming even without the incorporation of a robot.

When a robot is introduced, it can exacerbate this time burden. Holly (E1) explained: *"You know, I wasn't born and raised with technology like kids today. So, the technology end of it for me kind of felt like a lot of my focus in preparation"*. Therapists are therapy experts, but not necessarily experts in operating robots or other technologies. As Blanche (T2) puts it, *"I'm a therapist, like, I'm not like, like, I don't know tons of tech stuff, like, I am passable at best. ... I can troubleshoot small things, but when something massive is happening with an app that I know nothing about, I'm bewildered."* Holly (E1) mentions that *"it does require skill on the therapist"*. Even Isaac (N1), who suggested being technologically savvy (*"we're nerds here, and we like to do nerdy things"*), mentions that *"I took some practice because it was, it was a tedious process in the beginning."* Caroline (T3) shares that *"planning for that session (with a robot) for me was harder, really hard in the beginning"*, and that was because *"I did have to learn how to use the software."* Greg (T6) elaborates on the precise and time-consuming ways that the introduction of a robot exacerbates existing preparation challenges: *"you have to have your content, and your interface pretty well mastered ahead of time to be able to respond as quickly as you need to respond in a therapy session."*

Preparing robot content requires far more detail when compared to not using a robot. Caroline (T3), a therapist with over 5 years of experience, described how when using a robot, *"I actually have to plan even more."* They clarify: *"If it's just me talking I have control about myself. If I have to put things in the robot prior then he can only follow whatever I put there—my preparation needs to be well-polished."* Three of our participants (T1, T3, E1) explicitly described how planning details is a big part of preparing with a robot. Alissa (T1) mentions that with robots they had to *"pre-plan everything really extensively to be successful in those sessions."* When asked about what was involved in preparation with robots their initial response was just *"A lot. A lot."*

Here, notice:

1. Saad actually has three subsubthemes. Because he's at the penultimate level of his analysis, though, he breaks these into paragraphs instead of more strict subsubsections.
2. After listing each of his subsubthemes, Saad provides a few quotes that support that subsubtheme, working those quotes seamlessly into the rest of his writing.
3. Saad attributes each quote. You only have one interviewee so you can just say "Our interviewee" each time instead of providing a pseudonym and participant number.

Discussion

Next, add a third section called "Discussion" with two subsections: (1) Needs, (2) Recommendations.

Needs

To write the needs section, look as a group back through your writeup, and compile a list of (need-reason-source) triads based on your interview, each **need** is an identified need, each **reason** is your justification for that need, and each **source** is the axial code or corresponding rows in your transcript (or the articles) that back up that justification.

1. As a group, create a master list (re)defining your answers to the following questions (using a jamboard or similar to brainstorm any "new" questions)
 1. Your target application domain [Text A]
 2. The users you want to help within that domain [Text B].
 3. The most compelling needs your users have that you think a robot could help address: [Text C]
 1. Why each need is important to your target users, as justified by your axial codes: [Text D]
 2. Why you think this need **cannot** be satisfactorily addressed **without** a social robot: [Text E]
 3. Why you think this need **can** be satisfactorily addressed **with** a social robot: [Text F]
2. You're now ready to write your **vision statement**. Fill in the following Madlib, and then refine it as needed.

"Based on our results, our high level takeaway is that it is important for [Text B] in [Text A] to [Text C], in order to [Text D]. Unfortunately, [Text E]. We believe that Social Robots stand to address this problem: [Text F]."

Note that while this is in the discussion section of this writeup, it will be in the introduction of your final paper!

Recommendations

Finally, go back as a group through your writeup and brainstorm a list of recommendations based on your interview, that should guide the design of robots to meet the need listed above. Write them up as a series of design recommendations. Here is an example design recommendation writeup from Saad's paper:

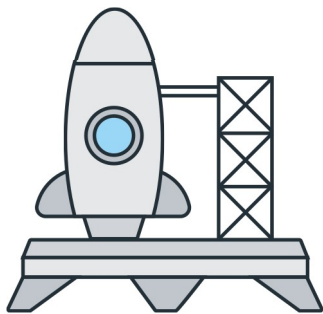
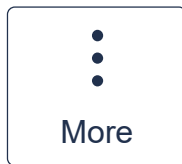
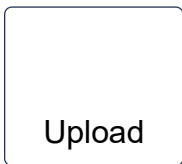
Recommendation: We recommend that developers building robot teleoperation tools present functionality for authoring content and operating robots in separate interfaces. Separating these modes of use acknowledges that authoring (preparing) and operating occur at different times and require different kinds of mental effort. Authoring generally occurs in the outer cycle when therapists have more time to be thorough. Teleoperation occurs during the more time-sensitive inner cycle of therapy sessions. Authoring interfaces ought to support planning for a high level of detail that simplifies teleoperation while incurring minimal inconvenience for the content author.

Deliverables

Submit to this assignment the report written across this lab, and your final coded interview.

Keep in mind, this submission will count for everyone in your Project Groups group.

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